Transforming Geo-Referenced Data in Contextual Information for Context-Aware Recommender Systems

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***Abstract*—A recommender system can be defined as an information filtering technology which can be used to output a ranking of items (e.g. products, places, etc) that are likely to be of interest to a user. Context-aware recommender systems makes recommendations by incorporating contextual information into the recommendation process. However, there is a lack of automatic methods to obtain contextual information for such systems. In this work, we have proposed to apply clustering techniques to transform geo-referenced data (i.e. latitude and longitude) in contextual information (i.e. regions) to feed the contextual systems. We have evaluated our proposal in the Yelp dataset, which showed evidences that our contextual information can provide better recommendations.**

***Index Terms*—Context-aware recommender systems, Geo-referenced data, Contextual information, Clustering**

I. Introduction

Due to the growth of the web and web-based systems, there was an increase in the amount of products and services that can be accessed by a user. With that amount of information, sometimes it is difficult for the users to choose the option that will fulfill their needs.

A recommender system can be defined as an information filtering technology which can be used to output a ranking of items (e.g. products, movies, places, etc) that are likely to be of interest to a user [19]. Recommender systems gather information on the preferences of their users for a set of items, being this information collected explicitly (by collecting user's ratings) or implicitly (by monitoring users behavior, such as songs heard, applications downloaded, web sites visited, books read, and so on) [6].

Traditionally, recommender systems consider only the interaction activity between users and items to provide recommendations. However, the use of contextual information can improve the recommendation process [2], [7], [16]. The researchers, that already investigated the use of contextual information, discovered that the quality of the recommendations increases when additional information, like time, location, and so on, are used.

In many applications, it may not be sufficient to use only information about users and items, it is also important to use contextual information in the recommendation process so that an item can be recommended only in certain circumstances. For example, a movie recommender system may use contextual information to recommend to the user movies to watch with its family or alone. As seen in [3], "Accurate prediction of consumer preferences undoubtedly depends upon the degree to which the recommender system has incorporated the relevant contextual information into a recommendation method".

Although the use of contextual information in recommender systems has received great focus in recent years [1], [2], [7], [10], [12], [16], there is a lack of automatic methods to obtain such information for context-aware recommender systems, specially in geo-referenced systems. In order to obtain these information, we have proposed to use three clustering techniques (i.e. DBSCAN, KMeans and Agglomerative Clustering) to transform geo-referenced data (i.e. latitude and longitude coordinates) in contextual regions (i.e. data clusters).

To evaluate our proposal, we have used CARSKit[1], a Java-Based Context-Aware Recommendation Library [27]. Besides the context-aware recommender systems already available in there, we added two others to this library, DaVI-Best [7] and Combined Reduction [2]. We have evaluated our proposal by combining the regions (i.e. the contextual information) with five different context-aware recommender systems to recommend places of interest to the users. We performed a set of experiments by using the Yelp dataset, which contains geo-reference data. The results showed evidences that the clustering of geo-referenced data in regions can provide better recommendations.

[1]https://github.com/irecsys/CARSKit

Thus, our main contributions in this work can be summarized as:

- Exploration of DBSCAN, KMeans and Agglomerative Clustering algorithms to transform geo-referenced data in contextual regions to improve context-aware recommender systems;
- Addition of two new context-aware recommender systems to the CARSKit library.

The rest of this paper is structured as follows: In section II we discuss some related works. The clustering algorithms used in this work are presented in Section III. The context-aware recommender systems used to evaluate our proposal are presented in Section IV. We evaluate our proposal in Section V. Finally, in Section VI, we present the conclusion and future work.

II. Related Work

In this section we discuss some recommender systems that make use of geo-referenced data. Two of the first recommender systems that use geo-referenced data are COMPASS [23] and LoCo [21]. These systems use the latitude and longitude points together with a radius value to calculate the area containing items which are suggested to be recommended by the systems.

Another geo-referenced recommender system is presented in [25]. Based on the location or the activity that the user desires, the system uses a collaborative filtering approach to generate the recommendations.

In [11], the authors present three Location-Aware Recommender Systems (LARS) that use the location of users, items or both to provide better recommendations. The LARS, that uses only items location, exploits a travel penalty which is based on the fact that users tend to travel a limited distance when searching for an item, and that for closer items, in travel distance to a user, it should be given precedence in the recommendation rank. This precedence is obtained by applying the travel penalty in the item score. In the rest of this paper, we will refer to this system only as LARS, and we will compare our proposal against it.

III. Clustering algorithms

We have exploited the DBSCAN, KMeans and Agglomerative Clustering algorithms to transform the geo-referenced data (i.e. latitude and longitude coordinates) in geographic regions (i.e. contextual information) as illustrated in Figure 1.

User	Place	Latitude	Longitude
u_1	i_1	37.806167	-122.448135
u_1	i_1	37.806348	-122.448355
u_1	i_1	37.806963	-122.448531
		...	
u_2	i_3	33.052551	-96.696394
u_2	i_3	33.020634	-96.698384

User	Place	Region
u_1	i_1	$region_1$
	...	
u_2	i_3	$region_3$

Fig. 1. Transforming geo-referenced data in geographic regions

A. DBSCAN

Density-Based Spatial Clustering of Applications with Noise (DBSCAN) is a data clustering algorithm which was proposed by [9]. The DBSCAN is a density-based clustering algorithm, which means that given a set of points in some space, it groups together points that are closely packed together (i.e. points with many nearby neighbors) and marks as outliers points that lie alone in low-density regions (i.e. points whose nearest neighbors are too far away).

For the purpose of clustering, the points are classified as core points, (density-) reachable points and outliers, as follows:

- A point p is defined as a core point if at least a number of points $MinPts$ are within of a distance Eps (Eps is the maximum radius of the neighborhood from p) from it (including p). All these points are said to be directly reachable from p;
- A point q is reachable from p if there is a path $p_1, ..., p_n$ with $p_1 = p$ and $p_n = q$, where each $p_i + 1$ is directly reachable from p_i. All the points on the path must be core points, with the possible exception of q;
- All other points not reachable from any other point are considered outliers.

Once we have classified the points, we can find the clusters. Each cluster will be composed by a core points p together with all other points (core or non-core) that are reachable from it.

B. KMeans

KMeans is a well-known partitional clustering algorithm that are often used due to its simplicity and efficiency. It has the parameter k, which represents the number of clusters that will be generated. Given a set of points, the algorithm iteratively splits the dataset in clusters based on a distance function, as for example, the Euclidean distance [13].

The algorithm, as seen in [15], is an iterative process that consists of two separate phases, in which the first one is responsible to select the k central points, called centroids. The second phase consist in assigning the remaining points to the nearest centroid. The algorithm is repeated until it satisfy a stop criteria. This criteria can be defined as follows [13]:

1) No reassignments of points to different clusters;
2) No change of centroids;
3) Minimum decrease in the Sum of Squared Error (SSE),

$$SSE = \sum_{b=1}^{k} \sum_{x \in C_b} dist(x, m_b)^2,$$

where k is the number of clusters, C_b is the cluster b, m_b is the cluster's b centroid and $dist(x, m_b)$ is the distance between the point x and the centroid m_b.

C. Agglomerative Clustering

The Agglomerative Clustering is the most well-known hierarchical clustering algorithm. The clustering process is made so that a nested sequence of clusters (similar to a tree) is built. The algorithm builds this tree from its leaves, where each leave is a point in the dataset, combining pairs of clusters that are similar or with the smallest distance between each other, repeating this process until that every point stays in the same cluster [13].

To measure the distance between clusters that can be combined, the average link method is commonly used, which calculates the mean distance between every point from a cluster to another [13].

IV. Context-Aware Recommender Systems

We evaluate our clusters/regions as contextual information in five context-aware recommender systems. Three algorithms that were already available in the CARSKit library (Item Splitting, User Splitting and UserItem Splitting) and two other algorithms that we added to the library (DaVI-Best and Combined Reduction).

These context-aware recommender systems are considered meta-algorithms, which means that they need an uncontextual recommender system to generate contextualized recommendations. In this work, we use the BPR algorithm [18] as this uncontextual system.

Personalized ranking aims at recommending items to the user in decreasing order of his/her preferences. In many situations, however, these preferences are only provided by implicit feedback from users. This means that the system only knows the positive observations; the non-observed user-item pairs can be either an actual negative feedback or simply the fact that the user does not know about the item's existence.

In this scenario, [18] proposed a generic method for learning models for personalized ranking. Instead of training the model using only the user-item pairs, they also consider the relative order between a pair of items, according to the user's preferences. Considering $N(u)$ to indicate the set of items for which user u provided an implicit feedback and $\bar{N}(u)$ to indicate the set of items that are unknown to user u, it is inferred that if an item i has been viewed by user u and j has not ($i \in N(u)$ and $j \in \bar{N}(u)$), then $i >_u j$, which means that he prefers i over j.

To estimate whether a user prefers an item over another, [18] proposed a technique denominated BPR (Bayesian Personalized Ranking) which is based on a Bayesian analysis using the likehood function for $p(i >_u j|\Theta)$ and the prior probability for the model parameter $p(\Theta)$. The final optimization criterion, BPR-Opt, is defined as:

$$\text{BPR-Opt} := \sum_{(u,i,j)\in D_K} \ln \sigma(\hat{s}_{uij}) - \Lambda_\Theta ||\Theta||^2 \quad , \qquad (1)$$

where $D_K = \{(u,i,j)|i \in N(u) \ \& \ j \in \bar{N}(u)\}$, $\hat{s}_{uij} := \hat{r}_{ui} - \hat{r}_{uj}$, r_{ui} is the known rating and $\hat{r}_{ui}$ is the predicted rating. The symbol Θ represents the parameters of the model, Λ_Θ is the corresponding set of regularization constants, and σ is the logistic function, defined as: $\sigma(x) = 1/(1 + e^{-x})$.

For learning the model, the authors also proposed a variation of the stochastic gradient descent technique, denominated LearnBPR, which randomly samples from D_K to adjust Θ. Algorithm 1 shows an overview of the algorithm, where γ is the learning rate.

Input: D_K
Output: Learned parameters Θ
Initialize Θ with random values
for *count = 1,...,#Iter* **do**
  draw (u, i, j) from D_K
  $\hat{s}_{uij} \leftarrow \hat{r}_{ui} - \hat{r}_{uj}$
  $\Theta \leftarrow \Theta + \gamma \left(\frac{e^{-\hat{s}_{uij}}}{1+e^{-\hat{s}_{uij}}} \cdot \frac{\partial}{\partial\Theta} \hat{s}_{uij} - \Lambda_\Theta \Theta \right)$
end

Algorithm 1: Learning through LearnBPR

A. Combined Reduction

In [2], the Combined Reduction approach uses the contextual information as label to segment the data. A segment is defined as a subset of the overall data selected according to the contextual information or combination of its values.

Briefly, this approach consists of the following two phases. First, using the training data, a recommendation method is run for each contextual segment (e.g. accesses on Mondays would be a segment) to determine which segment outperforms the traditional uncontextual recommendation model (i.e. using only user and item data). Second, taking into account the context of the user's active session, we choose the best contextual model to make the recommendation. Here the best model is the one with the highest F1 value obtained in the first phase of the approach.

B. DaVI-Best

In [7], we proposed a contextual modeling approach, called DaVI-Best, that treats contextual information as virtual items, using them along with the regular items in a recommender system.

Given a set of users U, a set of items I, a set of contextual values C and a set of multidimensional sessions S, where a session $s = \langle u, I, C\rangle$, the DaVI-Best approach consists of transforming each multidimensional session $s = \langle u, I, C\rangle$ into an extended two dimensional session $s' = \langle u, I \cup C\rangle$, where the values of the contextual information in C are used as virtual items along with the regular items in I. Once we have a set of extended two dimensional sessions S', building a contextual recommendation model consists of applying an uncontextual recommender algorithm on the set S'. Note that regular items are used to build the model and make recommendations. On the other hand, virtual items are used in addition to build/improve the recommendation model but they can not be recommended. We implemented a filter to guarantee this condition.

The DaVI-Best approach evaluates all contextual information and selects the one which better outperforms the traditional uncontextual recommendation model to make contextual recommendations.

C. Item Splitting

The Item Splitting approach, proposed by [5], focuses on the analysis of items with significant difference of ratings, in a same context, to split them into two contextual items according to the contextual information, as illustrated in Figure 2. Once we obtain the new data matrix, we can apply an uncontextual recommender system on it to provide contextualized recommendations.

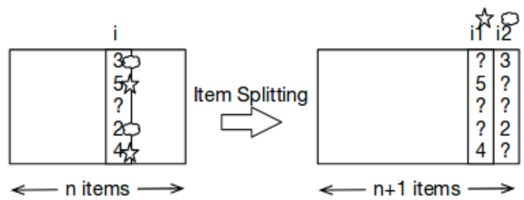

Fig. 2. Item Splitting approach [5]

D. User Splitting

The User Splitting [20] is an adaptation of the Item Splitting approach, where the user, in a given context, is analyzed and splitted in new users, as illustrated in Figure 3. Once we obtain the new data matrix, we apply an uncontextual recommender system on it to provide contextualized recommendations.

	u_i	u_j	u_k	u_m	u_j
i_a	1	3		5	
i_b		4			4
i_c			5	2	
i_d	5	3		3	
i_e	3	4	1		1

	u_i		u_j		u_k		u_m	u_l
	home	cinema	home	cinema	home	cinema	cinema	home
i_a	1			3			5	
i_b				4				4
i_c						5	2	
i_d	5		3				3	
i_e		3		4		1		1

Fig. 3. User Splitting approach [5]

E. UserItem Splitting

The UserItem Splitting unifies the Item and User splitting algorithms. Both item and user profiles are modified, generating a mix of them. This process is made by applying the Item Splitting followed by the User Splitting algorithm [26].

V. Empirical Evaluation

In order to transform the geo-referenced data in data regions, we use the scikit-learn[2] [17], a framework to build and evaluate machine learning and data analysis algorithms. The framework implements the three clustering algorithms used in this work: DBSCAN, KMeans and Agglomerative Clustering. To evaluate our proposal, we used the five context-aware recommender systems described in the previous sections. The algorithms are available in our extension of the CARSKit library[3].

A. Dataset

To evaluate our proposal, we used the dataset provided by the *Yelp Dataset Challenge*[4], and that after being preprocessed, contains 6,049 users, 5,414 items (i.e. places) and 19,986 ratings. Each item contains its geo-referenced data (i.e. latitude and longitude coordinates). The pre-processing consisted of removing unused attributes (i.e. date, hour and reviews) which were unrelated to our goal, and removing singletons from the dataset. Singletons are unique attribute values. Its removal produce better results in recommender systems, as seen in [8].

B. Experimental Setup and Evaluation Metrics

To measure the predictive ability of the recommender systems, we used the cross validation protocol, with 10-fold cross validation, and calculated the Precision, Recall and F1 metrics. In [22], the Precision metric is defined as the number of recommended items that are relevant, meanwhile the Recall metric is defined as the number of relevant items that are recommended. The F1 metric can be interpreted as the harmonic mean between the Precision and Recall values, with a range from 0 to 1. It is presented as:

$$\text{F1} = 2 * \frac{Precision * Recall}{Precision + Recall}.$$

For each configuration and metric, the 10-fold values were summarized by using mean and standard deviation. To compare two recommendation algorithms, we applied the two-sided paired t-test with a 95% confidence level [14].

C. Results

To obtain the results presented in this section, we have evaluated our proposal with different parameter setups in order to generate a diverse number of clusters/regions. In Table I, we present the parameters used by the clustering algorithms (the symbol "-" means that there is no setup), and the best results for each recommendation algorithm are presented in Tables II and III. In Table II, we compare our proposal against the BPR recommender system as baseline. In Table III, we compare our proposal against the LARS recommender as the baseline. Thus, we have evaluated our proposal against an uncontextual recommender system (BPR) and against a recommender system proposed in the literature that also use geo-referenced data (LARS). In Tables II and III, the values which are highlighted in boldface are the ones that are statistically significant.

The comparison against the baselines showed that it is possible to improve the recommendation process by using clusters/regions as contextual information. By comparing to

[2] http://scikit-learn.org

[3] https://drive.google.com/open?id=1dWpXFCFyt7uLqvqwBDpNGZZSzTg0IuoC

[4] https://www.yelp.com/dataset/challenge

TABLE I
CLUSTERING SETUPS FOR THE YELP DATASET

	KMeans	DBSCAN		Agglomerative
Setup	Num. Regions	Eps	MinPts	Num. Regions
Setup 1	2	0.001	1	2
Setup 2	8	0.001	5	8
Setup 3	16	-	-	16

the baselines, all three clustering techniques provided higher values of Precision, Recall and F1, that are statically significant, in special when the DaVI-Best recommender system is used.

In Table II, the DaVI-Best recommender presented the best performance with a F1 value of 0.0053570 for the KMeans algorithm. The setup that was used to obtain this result was the Setup 3, which generated 16 regions. By using the DBSCAN algorithm with the Setup 1, it was possible to generated 53 regions that, when evaluated with the DaVI-Best algorithm, it provided a F1 value of 0.0054110. For the Agglomerative algorithm, which generated the best F1 value among the three clustering algorithms, we have obtained a F1 value of 0.0055980 using 16 regions (Setup 3). Thus, the regions from the KMeans algorithm improved the F1 value by 49.14%, DBSCAN obtained an improvement of 50.64% and the Agglomerative Algorithm, which obtained the best performance, had an improvement of 55.85% against the BPR baseline.

As already stated, LARS is an algorithm that uses raw geo-referenced data to improve its recommendations. This algorithm outperforms the uncontextual BPR algorithm by 21.93% on F1, as can be seen by analyzing the results presented in Tables II and III. Even with this improvement over the BPR algorithm, regions as contextual information performed better than the raw geo-referenced data used by LARS.

When using LARS as a baseline, we obtain with KMeans an improvement of F1 by 19.66%, DBSCAN improved 20.86% and Agglomerative got a improvement of 25.04% in terms of F1 over the LARS algorithm.

VI. CONCLUSION AND FUTURE WORK

Geo-referenced systems can generate hundreds of data corresponding to the user coordinates. However, this huge number of observations is not useful if the objective is to represent where the user has been. In such a case, only a single data point is needed to represent the geographical location of the place that the user has visited. In order to obtain such a single data point to represent the user location, we have exploited the use of the several clustering algorithms to reduce the geo-referenced data (i.e. latitude and longitude coordinates) in data regions (i.e. data clusters). Then, we used the regions as contextual information in context-aware recommender systems. We evaluated our proposal in the Yelp dataset and the results showed that by clustering the geo-referenced data in regions, we can provide better recommendations.

As future work we can mention:

- Exploiting of the OPTICS algorithm instead of DBSCAN to generate density-based clusters [4];
- The replication of experiments using new datasets and other context-aware recommender systems;
- The combination of regions with other contextual information, for example, the time (i.e. hour) that the user was in the place, in order to provide better recommendations.

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TABLE II
COMPARING THE CONTEXTUAL ALGORITHMS AGAINST THE BPR BASELINE

Cluster. Method	Setup	Algorithm	Precision	Recall	F1
-	-	Baseline	0.0019760	0.0197570	0.0035920
Kmeans	Setup 1	**Combined Reduction**	**0.0025180**	**0.0208910**	**0.0044100**
Kmeans	Setup 3	**DaVI-Best**	**0.0030310**	**0.0265430**	**0.0053570**
Kmeans	Setup 1	Item Splitting	0.0027180	0.0224430	0.0047550
Kmeans	Setup 2	User Splitting	0.0025960	0.0217700	0.0045490
Kmeans	Setup 1	UserItem Splitting	0.0026350	0.0219480	0.0046070
DBSCAN	Setup 1	Combined Reduction	0.0020200	0.0166520	0.0035250
DBSCAN	Setup 1	**DaVI-Best**	**0.0030960**	**0.0258600**	**0.0054110**
DBSCAN	Setup 1	Item Splitting	0.0026130	0.0218930	0.0045870
DBSCAN	Setup 1	**User Splitting**	**0.0026640**	**0.0223490**	**0.0046650**
DBSCAN	Setup 1	**UserItem Splitting**	**0.0027560**	**0.0232720**	**0.0048390**
Agglomerative	Setup 1	Combined Reduction	0.0025010	0.0204920	0.0043610
Agglomerative	Setup 3	**DaVI-Best**	**0.0031860**	**0.0269810**	**0.0055980**
Agglomerative	Setup 3	Item Splitting	0.0025130	0.0213330	0.0044100
Agglomerative	Setup 1	**User Splitting**	**0.0026420**	**0.0215650**	**0.0046030**
Agglomerative	Setup 2	UserItem Splitting	0.0026200	0.0219140	0.0045880

TABLE III
COMPARING THE CONTEXTUAL ALGORITHMS AGAINST THE LARS BASELINE

Cluster. Method	Setup	Algorithm	Precision	Recall	F1
-	-	Baseline	0.0025460	0.0214800	0.0044770
Kmeans	Setup 1	Combined Reduction	0.0025180	0.0208910	0.0044100
Kmeans	Setup 3	**DaVI-Best**	**0.0030310**	**0.0265430**	**0.0053570**
Kmeans	Setup 1	Item Splitting	0.0027180	0.0224430	0.0047550
Kmeans	Setup 2	User Splitting	0.0025960	0.0217700	0.0045490
Kmeans	Setup 1	UserItem Splitting	0.0026350	0.0219480	0.0046070
DBSCAN	Setup 1	**Combined Reduction**	**0.0020200**	**0.0166520**	**0.0035250**
DBSCAN	Setup 1	**DaVI-Best**	**0.0030960**	**0.0258600**	**0.0054110**
DBSCAN	Setup 1	Item Splitting	0.0026130	0.0218930	0.0045870
DBSCAN	Setup 1	User Splitting	0.0026640	0.0223490	0.0046650
DBSCAN	Setup 1	UserItem Splitting	0.0027560	0.0232720	0.0048390
Agglomerative	Setup 1	Combined Reduction	0.0025010	0.0204920	0.0043610
Agglomerative	Setup 3	**DaVI-Best**	**0.0031860**	**0.0269810**	**0.0055980**
Agglomerative	Setup 3	Item Splitting	0.0025130	0.0213330	0.0044100
Agglomerative	Setup 1	User Splitting	0.0026420	0.0215650	0.0046030
Agglomerative	Setup 2	UserItem Splitting	0.0026200	0.0219140	0.0045880

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